

Lecture 30: LR Tests - Examples

Example 1. $x_1, \dots, x_n$ are iid observations from $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$; $\sigma > 0$

To test $H_0: \mu = \mu_0$ ag. $H_A: \mu \neq \mu_0$

(Recall, UMP(α) test does not exist in this case — graphically explained, UMPU(α) test exists, but we have not considered it, instead we proceed with the LR test)

Likelihood $f_{\underline{\mu}}$ under H_0 ,

$$L(\mu_0, \sigma^2 | \underline{x}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu_0)^2\right\}$$

$$\therefore \sup_{\sigma^2} L(\mu_0, \sigma^2 | \underline{x}) = (2\pi \hat{\sigma}_0^2)^{-n/2} \exp\left\{-\frac{1}{2\hat{\sigma}_0^2} \sum_{i=1}^n (x_i - \mu_0)^2\right\}, \quad \hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)^2$$

(usual) likelihood $f_{\underline{\mu}}$, $L(\mu, \sigma^2 | \underline{x}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right\}$

$$\wedge \sup_{\mu, \sigma^2} L(\mu, \sigma^2 | \underline{x}) = (2\pi \hat{\sigma}^2)^{-n/2} \exp\left\{-\frac{1}{2\hat{\sigma}^2} \sum_{i=1}^n (x_i - \hat{\mu})^2\right\}, \quad \text{where } \hat{\mu} = \bar{x}, \text{ and } \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Hence, the likelihood ratio criterion,

$$\lambda(\underline{x}) = \frac{\sup_{\sigma^2} L(\mu_0, \sigma^2 | \underline{x})}{\sup_{\mu, \sigma^2} L(\mu, \sigma^2 | \underline{x})} = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}^2}\right)^{-n/2} = \left(\frac{\hat{\sigma}_0^2 + n(\bar{x} - \mu_0)^2}{\hat{\sigma}^2}\right)^{-n/2} = \left(1 + \frac{t^2}{n-1}\right)^{-n/2},$$

$$\text{where } t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}} \quad (\text{as a r.v. this } t \sim t_{n-1}).$$

Then the critical region,

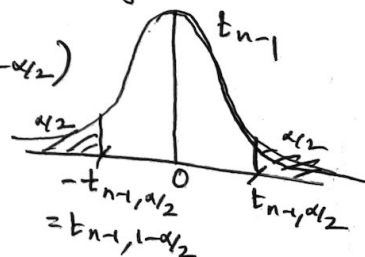
$$W: \lambda(\underline{x}) < c$$

$$\Leftrightarrow t^2 > c^* \Leftrightarrow |t| > c^{**}, \quad c^{**} \text{ obtained from size } \alpha \text{ condition, i.e.,}$$

$$P_{H_0}(|T| > c^{**}) = \alpha, \quad \text{use an equal-tailed test to get}$$

$$W: t > t_{n-1, \alpha/2} \quad \text{or} \quad t < -t_{n-1, \alpha/2} (= t_{n-1, 1-\alpha/2})$$

$$\text{i.e., } |t| > t_{n-1, \alpha/2}$$



Example 2 $X_1, \dots, X_n$ iid observations from $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$ known.
To test $H_0: \mu \leq \mu_0$ ag. $H_A: \mu > \mu_0$

(Recall: UMP(α) test exists for this case, but here we construct the LR(α) test & check whether the two leading decisions are same or different)

Under usual $\mathbb{P}$, mle of μ is $\bar{x}$.

$$\therefore \sup_{\mu \in \mathbb{P}} L(\mu | \bar{x}) = (2\pi\sigma^2)^{-n/2} \exp. \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right)$$

Under H_0 , mle of μ is, $\hat{\mu}_{mle} = \begin{cases} \bar{x} & \text{if } \bar{x} \leq \mu_0 \\ \mu_0 & \text{if } \bar{x} > \mu_0 \end{cases}$ [local problems on restricted parameter space, mle computation]

$$\text{Thus, } \sup_{\mu \in \mathbb{P}_0} L(\mu | \bar{x}) = \begin{cases} (2\pi\sigma^2)^{-n/2} \exp. \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2 \right\} & \text{if } \bar{x} \leq \mu_0 \\ (2\pi\sigma^2)^{-n/2} \exp. \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right\} & \text{if } \bar{x} > \mu_0. \end{cases}$$

Hence, the L.R. criterion μ

$$\lambda(\bar{x}) = \begin{cases} 1 & \text{if } \bar{x} \leq \mu_0 \\ \exp. \left[-\frac{1}{2\sigma^2} \left\{ \sum (x_i - \mu_0)^2 - \sum (x_i - \bar{x})^2 \right\} \right] & \text{if } \bar{x} > \mu_0. \end{cases}$$

and the critical region (from small $\lambda(\bar{x})$, only the 2nd line of above, if $\lambda(\bar{x}) = 1$, its max. value, H_0 is accepted).

$$\left\{ \bar{x}: \exp. -\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2 < c^* \right\}.$$

$$\text{i.e. } \left\{ \bar{x}: \bar{x} > \mu_0 + c^{**} \right\}, \quad c^{**} \text{ from size-}\alpha \text{ condn}$$

$$\text{i.e. } P_{H_0}(\bar{X} > \mu_0 + c^{**}) = \alpha \Rightarrow P_{H_0} \left(\frac{(\bar{X} - \mu_0)\sqrt{n}}{\sigma} > \frac{c^{**}\sqrt{n}}{\sigma} \right) = \alpha$$

$$\text{i.e. } 1 - \Phi \left(\frac{\sqrt{n}c^{**}}{\sigma} \right) = \alpha \Rightarrow \frac{\sqrt{n}c^{**}}{\sigma} = z_{1-\alpha} \text{ and } c^{**} = \frac{\sigma}{\sqrt{n}} z_{1-\alpha}$$

$$\text{Finally, } W: \bar{x} > \mu_0 + \frac{\sigma}{\sqrt{n}} z_{1-\alpha}$$

and the critical region is the same as the one obtained in the UMP(α) test!

